

Stability Analysis of the Two-Lumped App Model

Phys 39

App Model

Let T be the actuator-side temperature, and let T_m be the measured thermal-mass or thermistor temperature. The controller measures T_m , but the TEC first acts on T . With TEC power on and before PWM saturation, the Python app's mass mode is modeled as

$$\frac{dT}{dt} = h(T_{\text{room}} - T) + K_p(T_{\text{set}} - T_m), \quad (1)$$

$$\frac{dT_m}{dt} = m(T - T_m) + h(T_{\text{room}} - T_m). \quad (2)$$

The heat-loss coefficient h therefore appears in both thermal lumps: the actuator-side temperature T loses heat to room through $T_{\text{room}} - T$, and the measured mass T_m loses heat to room through $T_{\text{room}} - T_m$. The app slider $m = \text{mass}^{-1}$ is the rate at which T_m follows T ; smaller m means a slower measured mass.

Equilibrium

At equilibrium, $dT/dt = 0$ and $dT_m/dt = 0$, so

$$hT^* + K_p T_m^* = hT_{\text{room}} + K_p T_{\text{set}}, \quad (3)$$

$$-mT^* + (m + h)T_m^* = hT_{\text{room}}. \quad (4)$$

These two linear equations determine the steady temperatures. For

$$\beta = h(m + h) + mK_p,$$

the measured steady temperature is

$$T_m^* = \frac{h(m + h)T_{\text{room}} + mK_p T_{\text{set}}}{\beta}.$$

This is the proportional-control droop prediction for the measured temperature in the two-lumped app model.

Linear Perturbation Equations

Let

$$x = T - T^*, \quad y = T_m - T_m^*.$$

Constants such as T_{room} and T_{set} drop out of the perturbation equations:

$$\frac{dx}{dt} = -hx - K_p y, \quad (5)$$

$$\frac{dy}{dt} = mx - (m + h)y. \quad (6)$$

In matrix form,

$$\frac{d}{dt} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} -h & -K_p \\ m & -(m + h) \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}.$$

Characteristic Equation and Roots

Assume a mode proportional to e^{st} . The characteristic equation is

$$\det(A - sI) = 0.$$

For this model,

$$\det \begin{bmatrix} -h - s & -K_p \\ m & -(m + h) - s \end{bmatrix} = 0.$$

Expanding gives

$$s^2 + (m + 2h)s + [h(m + h) + mK_p] = 0.$$

Define

$$\alpha = m + 2h, \quad \beta = h(m + h) + mK_p.$$

Then

$$s^2 + \alpha s + \beta = 0,$$

with roots

$$s = \frac{-\alpha \pm \sqrt{\alpha^2 - 4\beta}}{2}.$$

Condition for Complex Roots

The roots become complex when the discriminant is negative:

$$\alpha^2 - 4\beta < 0.$$

Substituting $\alpha = m + 2h$ and $\beta = h(m + h) + mK_p$,

$$(m + 2h)^2 - 4[h(m + h) + mK_p] < 0.$$

The heat-loss terms cancel in the discriminant:

$$m^2 - 4mK_p < 0.$$

For $m > 0$, the complex-root condition is therefore

$$K_p > \frac{m}{4}.$$

Above this threshold, the roots are

$$s = -\frac{m + 2h}{2} \pm i \frac{\sqrt{4mK_p - m^2}}{2}.$$

Heat loss in both lumps does not move the complex-root threshold, but it does make the real part more negative:

$$\text{Re}(s) = -\frac{m + 2h}{2}.$$

Numerical Example from the App

Use the requested app parameters

$$T_{\text{set}} = 30^\circ\text{C}, \quad T_{\text{room}} = 20^\circ\text{C}, \quad K_p = 200, \quad h = 0, \quad m = 1.$$

The measured equilibrium temperature is

$$T_m^* = \frac{h(m+h)T_{\text{room}} + mK_p T_{\text{set}}}{h(m+h) + mK_p} = \frac{0 + 1 \cdot 200 \cdot 30}{0 + 1 \cdot 200} = 30^\circ\text{C}.$$

With these values,

$$\alpha = m + 2h = 1, \quad \beta = h(m+h) + mK_p = 200.$$

The characteristic equation is

$$s^2 + s + 200 = 0,$$

so

$$s = \frac{-1 \pm \sqrt{1 - 800}}{2} = -0.5 \pm 14.13 i.$$

The response is strongly underdamped because

$$K_p = 200 > \frac{m}{4} = 0.25.$$

The natural frequency, damping ratio, damped frequency, and oscillation period are

$$\omega_n = \sqrt{\beta} = \sqrt{200} = 14.14 \text{ s}^{-1},$$

$$\zeta = \frac{\alpha}{2\sqrt{\beta}} = \frac{1}{2\sqrt{200}} = 0.035,$$

$$\omega_d = 14.13 \text{ s}^{-1}, \quad T_{\text{osc}} = \frac{2\pi}{\omega_d} = 0.445 \text{ s}.$$

Because $K_p = 200$, the PWM saturates for errors larger than

$$\frac{255}{K_p} = 1.275^\circ\text{C}.$$

Thus this calculation describes small perturbations near equilibrium. A start from $T_m = 20^\circ\text{C}$ produces a 10°C error, so the early transient is governed by saturated PWM rather than this local linear model.

Continuous-Time Stability

For positive app parameters $m > 0$, $h \geq 0$, and $K_p > 0$,

$$\alpha = m + 2h > 0, \quad \beta = h(m+h) + mK_p > 0.$$

For a second-order continuous-time system,

$$s^2 + \alpha s + \beta = 0$$

is stable when both coefficients are positive. Therefore this ideal continuous two-lump model is stable for every positive proportional gain. It can still ring because the roots can be complex.

How to Promote Ringing

The controller input is T_m , while the controller output first acts on T . The path

$$\text{controller output} \longrightarrow T \longrightarrow T_m$$

introduces phase lag. To make ringing more visible in the ideal app model:

- Increase K_p : this moves the system farther above $K_p = m/4$.
- Decrease $m = \text{mass}^{-1}$: the measured mass responds more slowly, increasing the effect of the delayed feedback path.
- Decrease h : heat loss in both lumps makes the real part $-(m + 2h)/2$, so larger h damps the oscillation faster.

If the real apparatus shows sustained or growing oscillations, likely causes include sampling delay, thermistor lag, actuator lag, PWM saturation, quantization, or other nonlinearities beyond this continuous linear model.